

On Edge Coordination in Highly Dynamic Cyber-Physical Systems for Emergency Response

Amran Haroon, Mohammad Sagor, Maxwell Maurice[†], Liuyi Jin, Radu Stoleru, Roger Blalock[†]

Computer Science and Engineering, Texas A&M University, College Station, TX, USA

[†]Resilient Systems, NIST, Boulder, CO, USA

{amran.haroon, msagor, liuyi, stoleru}@tamu.edu, [†]{maxwell.maurice, roger.blalock}@nist.gov

Abstract—Mobile edge computing replicates cloud functions at the edge, thus reducing latency, improving responsiveness, and utilizing network bandwidth in resource-intensive applications such as Cyber-Physical Systems (CPS) for disaster response. These systems, which have gained significant interest from first responders, present immense potential to purvey the necessary computational power in highly dynamic and volatile environments like disaster response. However, existing edge computing solutions operate under the assumption that devices at the edge remain connected through a reliable edge network. While such systems bring enormous possibilities to first responders through new emergency response applications on handheld devices, they still suffer from poor inter-connectivity and, thus, remain ineffective. In this paper, we explore the impact of unreliable edge connectivity on edge computing functions, by deploying and evaluating a highly dynamic edge computing CPS consisting of an unmanned aerial vehicle (UAV) and mobile edge computing devices. We present our results from the deployment, showing limitations of current state-of-the-art solutions in tackling the challenges posed by unpredictable connectivity. We propose a solution for discovering services over intermittent LTE connections, to address some of the issues identified in the deployment.

I. INTRODUCTION

With the proliferation and advancements in hardware, from server clusters in data centers to personal computers, smartphones, wearables, and Internet of Things (IoT) devices, we are now in an information-centric era in which data is rapidly generated and stored on ubiquitous mobile devices [1]. To process such massive data, sending it over the Internet to the cloud, where powerful computation resources exist, is usually required. However, this is infeasible in scenarios, such as disaster response, where the Internet infrastructure is unavailable. The emerging edge computing paradigm addresses this challenge by collectively leveraging the far under-exploited resources on edge devices. In edge computing, several spatially close mobile edge devices form an edge cluster under *effective coordination*. Within a cluster, each mobile device is a service node that can appropriately share its underutilized resources (e.g., mobile CPU/GPU, communication, memory) while providing its own application services. In this way, data gets processed locally at the edge in the absence of communication with the cloud. This capability enables edge computing for critical Emergency Response Services (ERS) to first responders in a disaster area.

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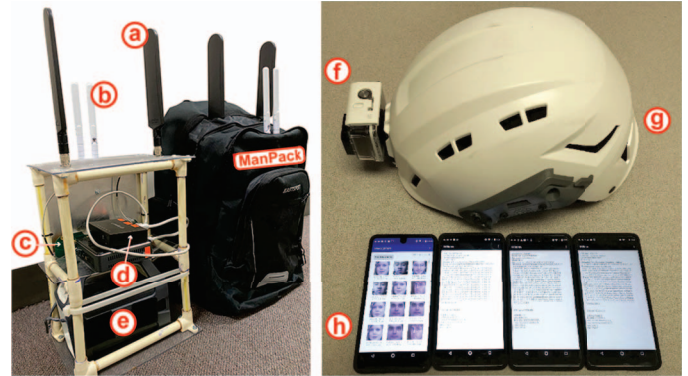


Fig. 1: DistressNet-NG edge-computing Cyber-Physical System consisting of a portable manpack (left), on-body camera and mobile devices (right), connected through 4G LTE and WiFi. The components include: a) LTE antenna, b) WiFi AP and Mesh, c) 4G LTE eNB, d) Intel NUC with High Performance Computing (HPC) capabilities, e) battery, f-g) on-helmet camera and h) mobile devices.

The devices employed in disaster response have, typically, diverse hardware configurations. Unmanned aerial vehicles may be present, wireless communication may be 4G LTE, WiFi or WiFi Direct, on-body cameras and other sensors may provide video streams for monitoring purposes while mobile phones mainly provide distributed computing resources. When these heterogeneous resources are *well coordinated* and the specialized neighboring services are quickly discovered, edge devices can rely on each other and work collaboratively to maintain effective ERS. These properties can be guaranteed given a reliably formed cluster with a suite of well-grounded functions (e.g., service discovery, leader election, edge configuration, status monitoring, etc.).

Despite these guarantees, the environment (e.g., location/altitude, landscape, obstacle orientation) in which mobile or aerial devices are deployed varies vastly, leading to highly dynamic signal attenuation and channel contentions. These dynamics introduce uncertainties to inter-connectivity among the mobile or aerial devices, which is detrimental to a *reliable edge formation* and consequently results in suboptimal ERS performance.

Cyber-Physical Systems (CPS) that integrate edge computing, resilient communication, and feedback control for drone-human coordination have recently received attention.



Fig. 2: DistressNet-NG deployments for *wildland firefighting in a large open field* in Gypsum, CO (2019) [shown in (a) and using the manpack shown in (b)]; and for *Urban Search and Rescue* in Disaster City, TX (2020) [shown in (c) and using (b) and the NIST rapidly deployable (d)]

DistressNet-NG [2], shown in Figure 1, is a CPS for Emergency Response. Designed for first responders, DistressNet-NG provides a distributed computation platform comprised of a scalable and resilient communication infrastructure. Through deployments in the lab and mobile environments (e.g., disaster responders performing wildfire-fighting or city-wide search and rescue, shown in Figure 2), DistressNet-NG has shown its potential to provide effective ERS [3]–[6]. However, still lacking are experiences of real system deployments in highly dynamic scenarios. Experiences such as drone-human interaction for disaster response can validate the current design decisions adopted in DistressNet-NG. To gain further insight into DistressNet-NG, we designed experiments for a highly dynamic scenario, deployed a UAV-assisted DistressNet-NG in an aerial field dedicated to UAV experiments, and evaluated it for 16 different use case scenarios. In this paper we: a) share our real-life experiences from deploying DistressNet-NG’s edge computing functions in a highly dynamic disaster response environment; b) share experimental results and their interpretation; c) and a first approach to address some of the issues identified in the deployment.

The rest of the paper is structured as follows. Section II presents background material and the motivation for this research. The system design for our deployment is described in Section III, the test results are presented in Section IV. We present a novel service discovery protocol for LTE networks in Section V and conclude in Section VI.

II. BACKGROUND AND MOTIVATION

In this section we present background material on DistressNet-NG and motivation for this research.

A. Background - DistressNet-NG

DistressNet-NG is a next-generation Cyber Physical System (CPS) that offers a scalable and resilient mobile edge computing platform designed for first responder mission-critical operations. DistressNet-NG enables distributed applications at the edge and employs mobile devices, high performance computing nodes (if available), on-body cameras and other sensors. It combines a state-of-the-art software stack and a deployable wireless communication infrastructure, which, in general, consists of the following 3 components (as shown in Figure 1): a) manpack (or rapidly deployable system) providing wireless

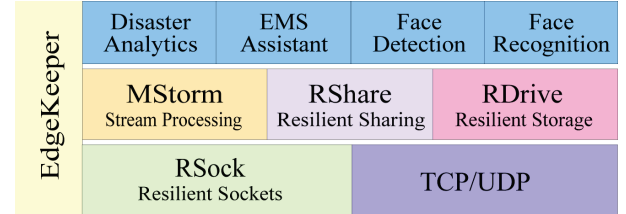


Fig. 3: DistressNet-NG edge computing software ecosystem

connectivity (WiFi, WiFi-Direct, 4G LTE) and edge computing capabilities (portable computer). The manpack holds the HPC node consisting an Intel NUC (32GB Intel Optane™ Module, 2TB HDD, 8GB DDR4-2400 SDRAM), a BaiCells Nova 227 4G LTE eNB (2496-2690MHz, maximum TX power 27dBm). Additionally, it has a Ubiquity EdgeRouterX, and a Unify 802.11AC Mesh radio, all powered by a Veracity PointSource Plus PoE injector for connecting the components. The 4G EPC (Evolved Packet Core) functionalities of the LTE are managed by the open-source NextEPC [7]; b) on-body cameras or other sensors; and c) mobile devices equipped with wireless communication capabilities (WiFi, WiFi-Direct, and 4G LTE) and on-board processing.

The DistressNet-NG software stack, shown in Figure 3, enables cloud-like processing (e.g., Hadoop [8] batch processing or Apache Storm [9] real-time stream processing) at the edge. It has five components: a) EdgeKeeper; b) RSock; c) RDrive; d) RShare; and e) MStorm. Among these components, EdgeKeeper and the Resilient Socket (RSock) are critical as all other components rely on them. EdgeKeeper, which runs on all devices of an edge cluster, is responsible for service discovery, device authentication, edge formation/reformation, edge health monitoring, topology management, metadata storage, etc. The distributed coordination capability enables EdgeKeeper to coalesce multiple mobile devices into one cluster while allowing devices to move from one edge to another. RSock provides opportunistic data delivery at the edge. It employs a Hybrid Routing Protocol (HRP) [10] that intelligently decides the degree of replication for a packet, to increase the data delivery probability. This is critical for resilient communication in an environment with sparse and highly dynamic wireless connectivity.

Apart from effective coordination and resilient commu-

nication, distributed stream processing along with resilient data storing and sharing capability is also essential for any dynamic CPS given unavailable cloud resources. Mobile Storm (MStorm [3]) is a stream processing platform that runs on mobile devices and provides the same real-time stream processing capability at the edge as Apache Storm provides in the cloud. RDrive in combination with RShare is a resilient and secure data storage and data sharing framework for mobile edge computing applications. RDrive employs encryption and erasure coding for security and resiliency against device failures. RSocket, MStorm, RDrive, and RShare exclusively rely on edge coordination services from EdgeKeeper. In a highly dynamic environment, resource-intensive ERS applications (e.g. face detection/recognition, bio-metric analysis, voice assistance, etc.) can only be successfully utilized by first responders with a state-of-the-art edge coordination service.

B. Motivation

EdgeKeeper's design employs a consensus protocol among multiple replica nodes for resilience, similar to ZooKeeper [11]. EdgeKeeper relies on a master node to discover other nodes in the network and selects suitable replicas before it can form an edge. With a replication factor of r , the master needs to find more than $r/2$ replicas in its neighborhood. The decision to assign service discovery functions to a master node was made due to the fact that LTE broadcast or multicast is not possible. Thus, service discovery protocols like mDNS, Avahi, and Bonjour are ineffective.

To assess initial design decisions, DistressNet-NG was evaluated in the lab and static outdoor environments, as shown in Figure 2. The system was deployed in Gypsum, CO, in a wildland-firefighting scenario (2019) and Disaster City, TX in an urban search and rescue scenario (2020). Those deployments were meant to evaluate EdgeKeeper's edge coordination capability in a real-life scenario with the evaluation metric being edge cluster formation latency. A shorter edge formation latency can make better use of the sparse, and possibly, short periods of connectivity at the edge. Some of our experimental results are shown in Figure 4. We can observe that both in the lab and static outdoor environments, increasing the number of replicas the edge formation latency increases considerably, from a few seconds, for one replica, to up to 25sec for a 3 or 5 replica configuration.

While these results show the coordination capability of EdgeKeeper when DistressNet-NG is deployed in a real-world environment, it is important to note that the deployed devices were relatively close to each other ($\leq 60\text{m}$) and that neither the lab nor the outdoor environments were highly dynamic. The introduction of an aerial device (e.g., UAV) can provide such dynamics, similar to what first responders usually face at disaster scenes [12]. Consequently, the research questions we were interested in answering were: *What is the impact of a highly dynamic environment on EdgeKeeper's edge formation and reformation latency? What is the impact of a highly dynamic environment on DistressNet-NG's edge computing applications that rely on EdgeKeeper for edge coordination?*

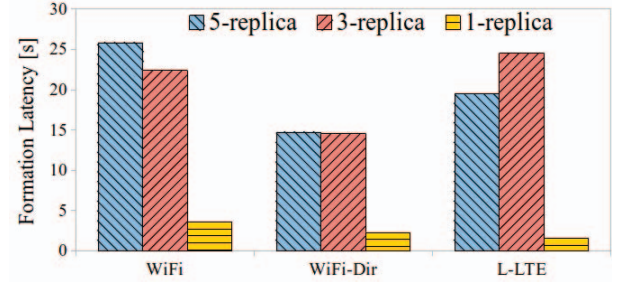


Fig. 4: EdgeKeeper edge formation latency in lab/static environments, with different wireless technologies (WiFi, WiFi-Direct, and Lab LTE (L-LTE)) and with different degrees of fault tolerance (1-5 replicas)

III. SYSTEM DEPLOYMENT

In this section, we present the details of DistressNet-NG's UAV-assisted real-world deployment. In our experiment, we use the UAV as an aid to form an edge computing cluster with mobile devices on the ground. Our deployment would simulate a larger scale disaster response where mobile devices on the ground would not be able to collaborate for accomplishing a mission, without the aid of a UAV (which provides wireless connectivity and computation resources).

All experiments were conducted between February 25-28, 2021 at the Christman Airfield in Fort Collins, CO, a premier site for UAV testing, flight, training, and research. Figure 5 depicts time instances of the deployment: the tripods used for placing the mobile devices in the field (Figure 5a), the UAV carrying the eNodeB with a MasterNode (Figure 5b), and the station used for system setup and troubleshooting (Figure 5c).

The UAV was a DJI M600 Pro multi-rotor drone (Figure 5b) with a payload of a FeatherLite [13] LTE eNodeB (locked on LTE Band 14) and a mobile device (Samsung Galaxy S9) configured as Master Node with the DistressNet-NG software stack. We employed an additional 12 mobile devices (Samsung Galaxy S9 and Sonim XP8), all running DistressNet-NG. Six additional Bittium Tough phones were used for logging LTE Reference Signal Received Power (RSRP).

For our system deployment, we were interested in evaluating the *edge formation and replica selection (reformation) latency* for EdgeKeeper, the *task completion rate* for MStorm, the *data delivery success rate* for RShare, and the *data read/write success rate* for RDrive. We were also interested in recording the raw wireless link quality (RSRP) and trying to correlate it with application performance.

For this, we designed a deployment with an architecture shown in Figure 6a. The UAV would start its flight from a central position (C in Figure 6b) and fly in a circular fashion on two different orbits (radii of 350m and 600m) at an approximate altitude of 122m. The devices on the ground were stationary, placed on individual tripods, away from the center point of the orbit, at locations A through D in Figures 6b-6c, depending on the test scenario. In some scenarios multiple mobile devices were placed at the same location, to simulate a group of collocated disaster responders.

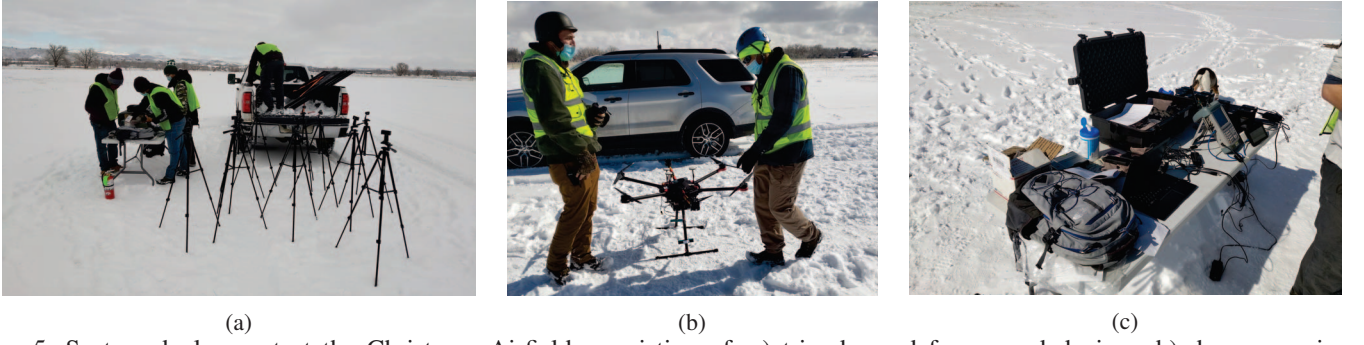


Fig. 5: System deployment at the Christman Airfield, consisting of: a) tripods used for ground devices; b) drone carrying eNodeB and Master Node mobile device; and c) setup station for system initialization and troubleshooting

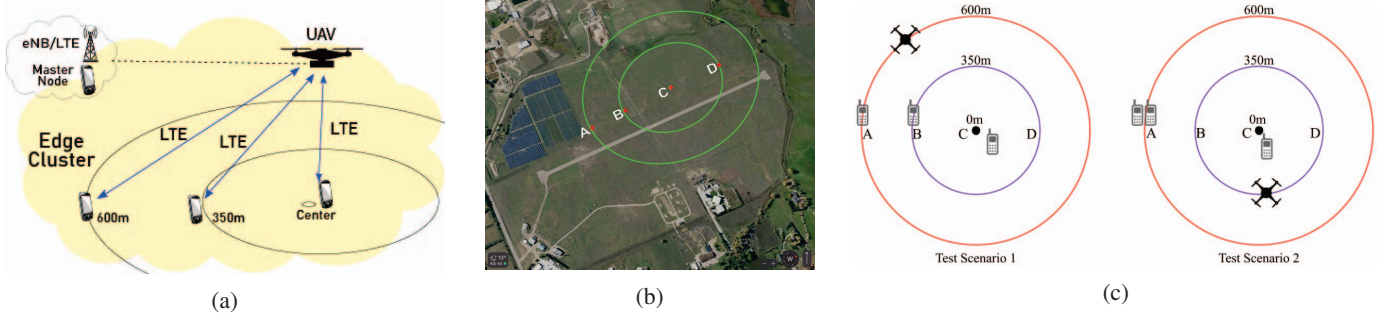


Fig. 6: Deployment design, location and test scenarios: a) deployment architecture; b) UAV flight paths and mobile device locations superimposed on the map of Christman Airfield, CO; c) two deployment scenarios where the UAV aims to aid mobile devices on the ground establish an edge computing cluster

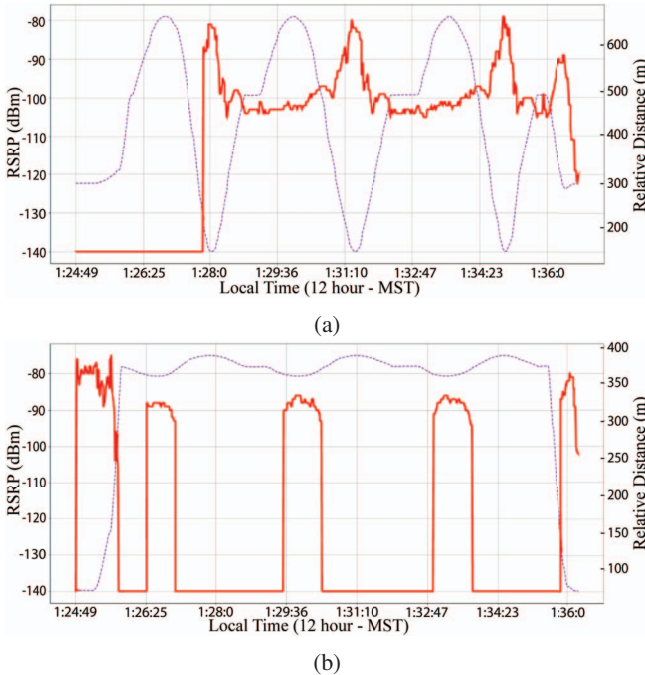


Fig. 7: RSRP as a function of the relative distance between the UAV and mobile device when: a) mobile logging device is at *D*; and b) mobile logging device is at *C*

IV. EXPERIMENTAL RESULTS

A total of 8 different flight plans involving 16 different test scenarios were used. Two such scenarios are shown in Figure 6c where the UAV was flying in circles with a 600m

radius (Test Scenario 1) and with a 350m radius (Test Scenario 2). Every flight had a 20min duration on average. During the 20min window, the UAV could complete approximately 5 orbits at 350m and 3 orbits at 600m radius.

In this section, we present experimental results from our UAV-assisted deployment and compare them with results from previous deployments (that also served as motivation for this research and were presented in Section II).

A. UAV 4G LTE-based Edge Connectivity

From our deployment, we have collected a large wireless connectivity dataset. Some representative results are shown in Figure 7, which depicts the RSRP (as recorded at locations *A-D*) as a function of the distance between the UAV and the logging location.

The result in Figure 7a is when the logging device is at *D* and the UAV flies on a 350m orbit. As the figure shows, the initial relative distance between the UAV and device (UAV-device) is about 300m. While the UAV departs towards location *B* the UAV-logger relative distance increases and the RSRP never reaches a level high enough for connectivity. The first time when a sufficiently high RSRP is recorded is when the UAV is above location *D*, thus at the shortest UAV-logger relative distance (the distance is only due to the altitude of the UAV). Interestingly, the RSRP level remains high enough even after the UAV-logger relative distance increases again, and only decreases when the UAV returns to the ground at the end of the experiment.

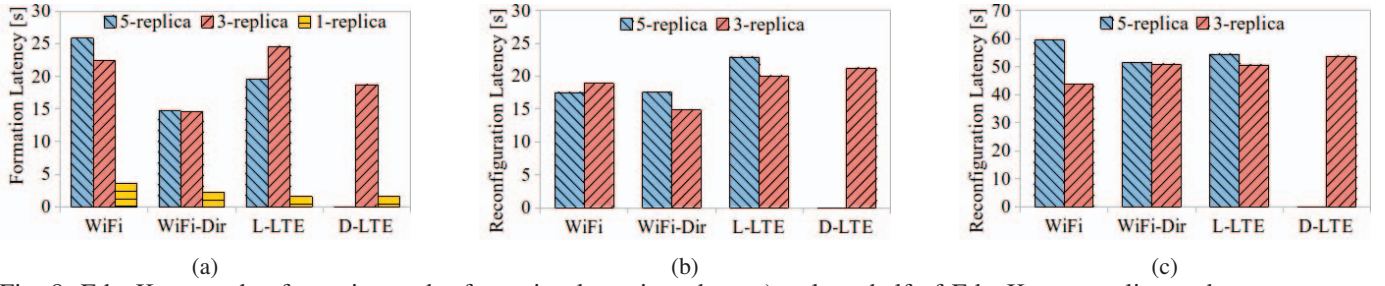


Fig. 8: EdgeKeeper edge formation and reformation latencies when: a) at least half of EdgeKeeper replica nodes are present; b) one node joins an existing cluster and it is selected as replica; and c) one replica node leaves the edge cluster, resulting in no consensus possible and requiring an edge reformation

The result in Figure 7b is when the logging device is at position *C* and the UAV flies on a 350m orbit. As shown, the relative distance UAV-logger remains largely at 350m, with slight variations only due to imperfect circular UAV orbit, when compared with the logger on the ground. Interestingly, the RSRP varies significantly and only reaches high enough levels (e.g., -80dBm) when the distance is relatively shorter. Just a few tens of meters difference in the relative distance made the UAV lose connectivity with ground devices. In some of these experiments, we observed 3 nearby AT&T cell sites (operating also on Band 14), with RSRP of -95 dBm (2 sites) or -105 dBm (1 site). This could have been a major reason for the observed phenomena and is consistent with other interference measurements [12].

B. Edge Formation and Reformation Latency

The experimental result shown in Figure 7b illustrates the importance of agile edge formation, as opportunities for edge computing in highly dynamic scenarios may be seldom and short-lived (e.g., 60-120sec time periods of connectivity in Figure 7b). A natural question is how our highly dynamic environment impacts the edge formation and reformation latency (motivated in Section II and illustrated in Figure 4).

The results from our deployment are presented in Figure 8, where D-LTE represents the results from our current deployment, presented in comparison with the lab/static results previously presented in Figure 4. As shown in Figure 8a, the edge formation delay in our deployment is close to the results obtained in the lab/static environment, for both 1 and 3 replicas (we did not experiment with 5 replicas using D-LTE due to expected long edge formation delay and the expected short periods of connectivity). As shown, for 1 replica edge coordination, an edge formation latency of under 5sec is possible. Yet, for 3 or 5 replica scenario, latencies of 15-20sec were observed.

The experimental results for the scenario where one node joins an existing edge cluster and it is selected as a replica, are shown in Figure 8b. We note that the edge reformation latency slightly increases by about 5sec when compared with the initial edge formation latency. It is important to remark that the D-LTE results obtained in this deployment are largely similar to the lab/static environments (WiFi, WiFi-Dir, and L-LTE).

Figure 8c depicts the edge reformation delay, when one replica of EdgeKeeper departs. This scenario is more difficult to handle since the detection of a replica's departure cannot be easily distinguished from a slowly replying replica (due to communication or computation latencies). The experimental results show that edge reformation latencies increase across all wireless technologies, to 45-60sec. Equally important to observe is the fact that the reformation latency for the highly dynamic environment (D-LTE) is similar to the other environments (WiFi, WiFi-Dir, and L-LTE).

The presented edge formation and reformation latencies, however, do not depict a complete picture, as edge computing application performance is what DistressNet-NG aims to provide for disaster response scenarios. To investigate this, we conducted a set of experiments described below.

C. Impact of Edge Coordination on Edge Computing Applications - MStorm, RShare and RDrive

For these experiments we selected three runs that used Test Scenario 2 in Figure 6c: the UAV was flown on a 350m orbit and ground devices were placed at positions *A* and *C*. The results for the MStorm, RShare, and RDrive success rates are depicted in Figure 9. We can see that for all applications, the completion rate is less than 100%.

The task completion rate for MStorm (shown in Figure 9a) varies between 20% to 70%. In these experiments, we observed a multitude of edge disconnection events. Due to the lengthy edge reformation latency and the reliance on a single master node to form an edge, it was frequently the case that the edge formed successfully, the MStorm application started, yet an edge disconnection event rendered MStorm ineffective, not being able to complete task computation.

Similar results were obtained for RShare (shown in Figure 9b), with data delivery rates between 40% to 90%. Similar observations to MStorm, namely edge disconnection events, followed by lengthy edge reformation delays, were found to be the culprit for the performance.

The results obtained from RDrive experiments (shown in Figure 9c) show the difficulty of edge formation was in the UAV-based deployment. The RDrive application employed a series of write operations followed by reads of the same data. While the success rate of RDrive varies from failure to 85%, it is important to remark that the ratio Write/Read

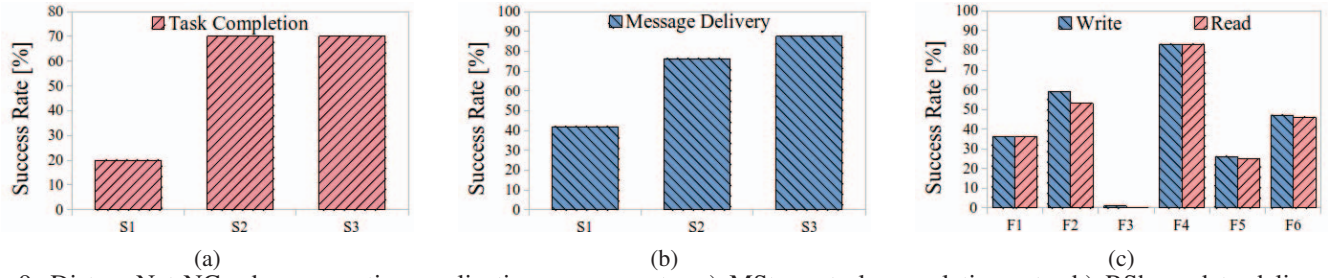


Fig. 9: DistressNet-NG edge computing application success rate: a) MStorm task completion rate; b) RShare data delivery success rate; and c) RDrive data storage write/read success rate

TABLE I: JmDNS+ WiFi and LTE service discovery latency

Wireless Link	Discovery Latency [sec]
WiFi	2.23
LTE	2.68

is frequently close to 100%, emphasizing that if the edge is stable, without disconnections, and if the write operations complete successfully, then it is very likely that the read operations complete successfully as well.

These experimental results served as the motivation for addressing the edge formation and reformation latency and for removing the dependence on a master node. Our solution is presented in the following section.

V. SERVICE DISCOVERY FOR HIGHLY DYNAMIC EDGES

Existing service discovery protocols, Avahi [14], Bonjour [15], NSD (Android) and JmDNS [16], employ DNS-SD (DNS Service Discovery) or mDNS (multicast DNS) for service discovery and registration. The implementation of DNS-SD and mDNS require support for multicast/broadcast, capability that is not available over 4G LTE wireless links. Thus, the aforementioned initial design decision was made, to employ a master node in DistressNet-NG.

To address this limitation, we extended the capability of JmDNS. In our solution, which we named JmDNS+ performs unicast based service discovery over LTE by periodically sending unicast packets to all IP addresses in the class. Devices running JmDNS+ register and announce themselves upon receiving of a unicast message over LTE. The registration and announcement over WiFi was kept intact to that of JmDNS, by employing multicast. The size of the LTE unicast packet is relatively small, typically less than 100B. JmDNS+ retains the same interface as JmDNS, hence remains fully compatible with all zero configuration protocols over WiFi and Ethernet.

The performance evaluation of JmDNS+, shown in Table I, indicates that service discovery over any wireless link (WiFi or LTE) is fast, within a few seconds. This, coupled with an edge leader election protocol can enable a fast edge formation and reformation process and removes the dependency on a single master node.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, we present a real-world deployment experience of DistressNet-NG (a state-of-the-art, scalable, and resilient mobile edge computing platform designed for first responders) in a highly dynamic 4G LTE-only UAV-assisted

environment. Experiments were conducted for service discovery/coordination, edge formation/reformation latencies, data storage, and stream processing at the edge. These illustrated the importance of edge coordination for edge computing applications. To address the challenges observed, a fast service discovery primitive for diverse wireless links (LTE and WiFi) was developed. For future work, we leave the evaluation of the faster service discovery primitive within the entire DistressNet-NG ecosystem in an emulation environment [4] by replaying collected wireless connectivity datasets.

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